A Formal Skeleton for a Universal Symbolic System

Abstract

We specify a finite symbolic calculus, a coding layer over binary patterns, a resonance functional on coded naturals, and a fixed-point learning operator with existence guarantees. All claims are stated with explicit hypotheses. Prime-related statements are framed as propositions or conjectures with defined evaluation procedures.

1. Logical Signature and Semantics

Signature. Work in single-sorted first-order logic with equality over signature

$$\Sigma = \{0, S, +, \cdot, \leq, \mathsf{if}, \mathsf{pair}, \mathsf{fst}, \mathsf{snd}\}.$$

Axioms. Let **A** be Peano Arithmetic (PA) in the language $\{0, S, +, \cdot, \leq\}$. The auxiliary constructors $\mathsf{if}, \mathsf{pair}, \mathsf{fst}, \mathsf{snd}$ are defined by conservative definitional extensions inside PA. All reasoning about naturals occurs in models of $(\Sigma, \mathbf{A})$.

Remark. "Universal" means: any computable structure over a finite alphabet admits an interpretation in some model of $(\Sigma, \mathbf{A})$ via standard encodings.

2. Alphabet and Syntax Layer

Let the concrete alphabet be

$$\mathcal{A} = \{\mathtt{0}, \mathtt{1}, \mathtt{S}, +, \backslash *, (,),, \}.$$

Terms and formulas follow standard first-order formation rules. Programs are primitive-recursive schemata definable in PA. No minimality of $|\mathcal{A}|$ is claimed.

3. Coding Layer over Binary Patterns

Fix an integer $M \geq 1$. Write the binary cube as $\{0, 1\}^M$, identified with the vector space $\mathbb{F}_2^M$.

Coding of naturals. Choose a computable injective map $c_M : \mathbb{N} \to \{0, 1\}^M$. Two canonical choices: 1. **Bit-slice:** $c_M(n)$ is the length-M binary expansion of $n \bmod 2^M$. 2. **Prime-exponent slice:** Let $p_1, \ldots, p_M$ be the first M primes. For $n = \prod_i p_i^{e_i} r$ with $\gcd(r, \prod_i p_i) = 1$, set $c_M(n)_i = e_i \bmod 2$.

Fix one choice and use it consistently. For asymptotics, consider the directed system as $M \to \infty$ with cylinder projections.

4. Resonance Functional

Let the open simplex be

$$\Delta_M = \{\alpha \in (0,\infty)^M : \sum_{i=1}^{M} \alpha_i = 1\}.$$

Define, for $b \in \{0,1\}^M$ and $\alpha \in \Delta_M$, the multiplicative score

$$f_\alpha(b) = \prod_{i=1}^{M} \alpha_i^{b_i}.$$

For $n \in \mathbb{N}$, write $b(n) = c_M(n)$ and set $s_\alpha(n) = \log f_\alpha(b(n)) = \sum_{i=1}^{M} b(n)_i \log \alpha_i$.

Local comparison. Fix a finite, undirected graph $G = (\mathbb{N}, E)$ with bounded degree; typical choices: - **Adjacency-1:** $E = \{\{n, n \pm 1\}\}$. - **Residue-neighbors:** connect n to $n \pm p_i$ for $i \le M$.

Define the neighborhood mean $\mu_G(n) = \frac{1}{\deg(n)} \sum_{m \sim n} s_\alpha(m)$, and the anomaly

$$r_\alpha(n) = s_\alpha(n) - \mu_G(n).$$

5. Fixed-Point Learning Operator

Let $K = \Delta_M \times [\underline{\tau}, \overline{\tau}] \subset \mathbb{R}^{M+1}$ with $0 < \underline{\tau} < \overline{\tau} < \infty$. Define a continuous operator

$$F(\alpha,\tau) = \Big(\Pi_{\Delta_M}\big(\alpha + \eta\, \nabla_\alpha \Phi(\alpha,\tau)\big),\ \Pi_{[\underline{\tau},\overline{\tau}]}\big(\tau + \eta\, \partial_\tau \Phi(\alpha,\tau)\big)\Big),$$

where Π are Euclidean projections, $\eta > 0$, and

$$\Phi(\alpha,\tau) = -\mathbb{E}_{n \le N}\big[r_\alpha(n)^2\big] \quad \text{or} \quad \Phi(\alpha,\tau) = H(\alpha) - \lambda\, \mathbb{E}_{n \le N}\big[r_\alpha(n)^2\big]$$

with entropy $H(\alpha) = -\sum_i \alpha_i \log \alpha_i$ and constants $\lambda > 0, N \in \mathbb{N}$.

Theorem 5.1 (Existence). K is compact and convex; $F : K \to K$ is continuous. Hence F admits a fixed point by Brouwer.

Sketch. Projections are continuous. Φ is continuous on the compact set. Composition preserves continuity. Brouwer applies.

6. Variational Problem for Parameters

Consider the convex optimization

$$J(\alpha) = \mathbb{E}_{n \le N}\big[r_\alpha(n)^2\big] \quad \text{s.t.} \quad \alpha \in \Delta_M.$$

Proposition 6.1 (Existence of minimizers). J is continuous on compact Δ_M ; a minimizer α^* exists by Weierstrass. Uniqueness requires strict convexity, which holds if the neighborhood statistics induce a strictly positive-definite covariance of $b(n)$ under the chosen G and sampling.

7. Primality-Related Statements

Let $\mathsf{Prime}(n)$ denote primality in PA.

Definition 7.1 (Resonance sieve score). For a fixed (M, G, α) , define the score $\rho(n) = -r_\alpha(n)$. Smaller ρ indicates local under-representation of the pattern bits of $n)$ relative to neighbors.

Claim Type A (Decision rule). For a threshold θ , define $\mathcal{P}_\theta = \{n \le N : \rho(n) \le \theta\}$. This is an algorithmic filter.

Proposition 7.2 (False-positive control, finite model). Fix (M, G, α) . For uniform sampling over $[1, N]$ and any threshold θ , the empirical false-positive rate of $\mathcal{P}_\theta$ with respect to Prime admits Chernoff-Hoeffding bounds computable from $\{r_\alpha(n)\}_{n \le N}$. No logical equivalence with Prime is asserted.

Conjecture 7.3 (Structured anomaly at primes). For certain (M, G) and α^* minimizing J , the distribution of $r_{\alpha^*}(n)$ places primes in the lower tail relative to composites at growing scales. This is empirical and requires calibration; it is not equivalent to primality.

8. Cycles and Automata

Let C_k be the cyclic group of order k . A length-k cycle in the coding layer is a homomorphism $\varphi : C_k \to \{0,1\}^M$ under componentwise XOR. Families of cycles are described by subgroup data of $\mathbb{F}_2^M$. Counts like 48/96/256 refer to group orders and direct sums; they are not asserted as universal constants.

9. Artifact Algebra and a Conserved Quantity

Let $(\mathcal{X}, \oplus)$ be a free commutative monoid generated by primitive artifacts. Define a degree map $\deg : \mathcal{X} \to \mathbb{N}$ by $\deg(x \oplus y) = \deg(x) + \deg(y)$ and $\deg(\mathbf{0}) = 0$. For any endomorphism $T : \mathcal{X} \to \mathcal{X}$ that is a monoid homomorphism, $\deg$ is conserved: $\deg(T(x)) = \deg(x)$.

10. E_8 Reference Construction (Optional)

Define the lattice

$$E_8 = \left\{x \in \mathbb{Z}^8 \cup \left(\mathbb{Z} + \tfrac{1}{2}\right)^8 : \sum_{i=1}^{8} x_i \equiv 0 \pmod 2\right\},$$

with bilinear form $\langle x, y \rangle = \sum_i x_i y_i$. The 240 roots are the vectors of squared norm 2. A coding map $\gamma : \{0,1\}^8 \to E_8$ may be chosen via a Gray map followed by centering. No structural equivalence between the resonance system and E_8 is claimed here.

11. Computation and Evaluation Protocol

1. Fix (M, G) and training window $[1, N]$.
2. Compute $\alpha^* \in \arg\min_{\Delta_M} J(\alpha)$.
3. Evaluate $r_{\alpha^*}(n)$ on a disjoint test window $[N+1, 2N]$.
4. Choose threshold θ by ROC analysis against ground-truth primality.
5. Report precision/recall and calibration curves with concentration bounds.

12. Theorems and Their Dependence

- Existence of fixed points (Thm 5.1) depends only on compactness, convexity, and continuity.
- Existence of minimizers (Prop 6.1) depends on continuity over compact Δ_M .
- Any stronger claims relating r_α to Prime are empirical or conjectural.

13. Summary of Objects

- Logical base: PA over Σ .
- Coding: injective $c_M : \mathbb{N} \to \{0,1\}^M$.
- Parameters: $\alpha \in \Delta_M$, graph G with bounded degree.
- Scores: $s_\alpha(n)$, anomalies $r_\alpha(n)$.
- Operators: gradient-projected F with a Brouwer fixed point.
- Evaluation: finite-sample statistical testing; no equivalence with primality asserted.

14. Implementation Notes

- All definitions are internal to PA with standard encodings.
- For numerical work, use M large enough for the application, and confirm stability of α^* under increasing M .
- Use disjoint train/test ranges to avoid leakage in r_α statistics.

15. Open Problems

1. Characterize (M, G) for which J is strictly convex.
2. Prove or refute Conjecture 7.3 under plausible randomness models for composites.
3. Determine whether any nontrivial algebraic structure links cycle families to classical lattices beyond coding-theoretic embeddings.